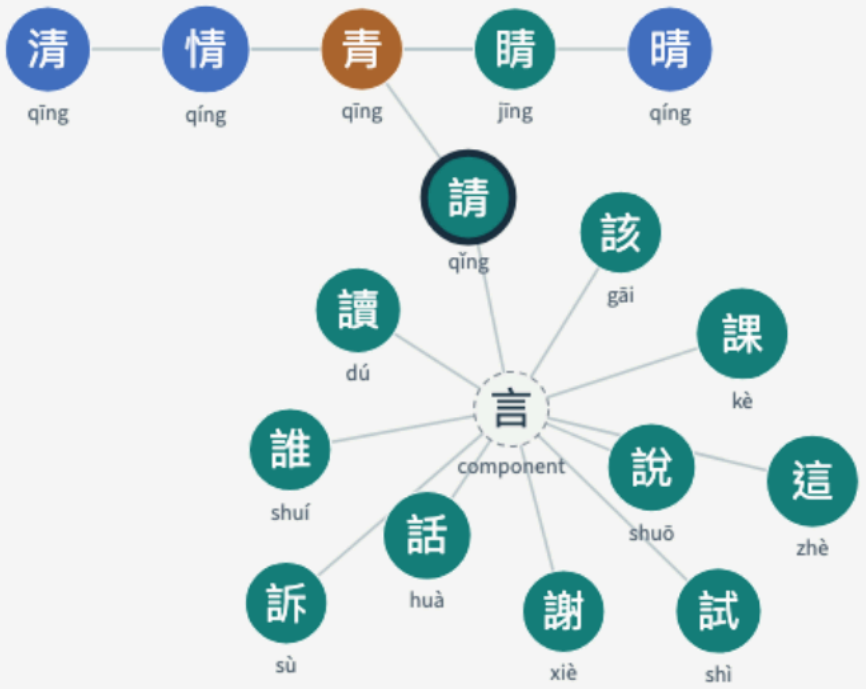
ZIKEY

Seeing and steering your next character

Self-study offers long vocabulary lists. Learners need connected steps and reasons they can inspect.



A working interactive prototype

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Real data in a bounded pilot

290 characters **450 vocabulary entries** **556 graph edges**

Source	Fields used	Snapshot / attribution
TOCFL	Traditional words, pinyin, level	Workbook dated 2024-09-23
Make Me a Hanzi	Components, radical, origin hints	LGPL v3+ dictionary
CC-CEDICT	English word meanings	2026-11-04 / CC BY-SA 4.0

6 missing English glosses / 30 missing origin records

Missing fields stay visible. Vocabulary prevalence serves as a labelled proxy for corpus frequency.

Local system and data flow

01 **Preserved source snapshots**

Process once into characters, words and leaf-component edges.

02 **Inspectable recommendation**

Compute four signals from the current known-character set.

03 **Three linked views**

D3 network, score breakdown and Plotly coverage comparison.

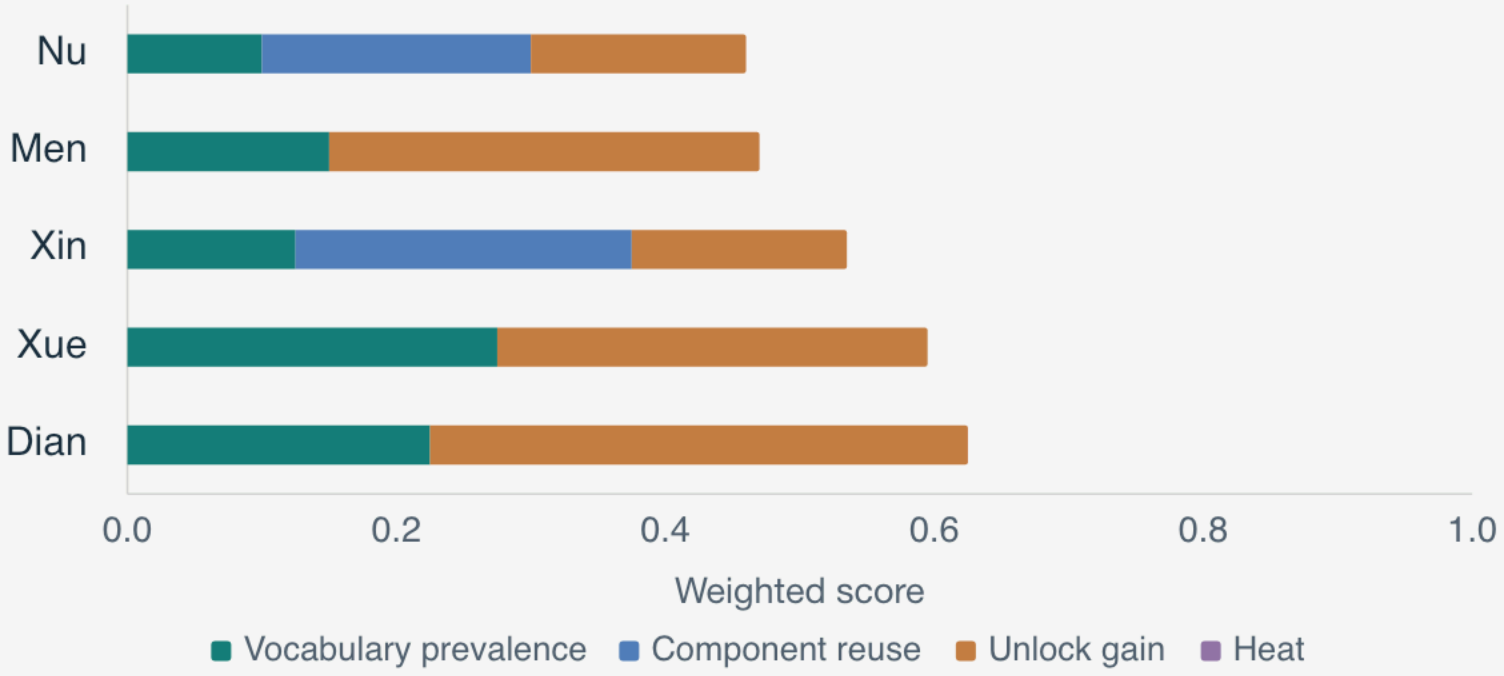
04 **On-device explanation**

Ollama selects up to two supported evidence sentences.

One shared selection and known-character set connect the entire page.

Four inspectable scoring signals

Default weights: prevalence 35%, reuse 25%, unlock 40%, heat 0%



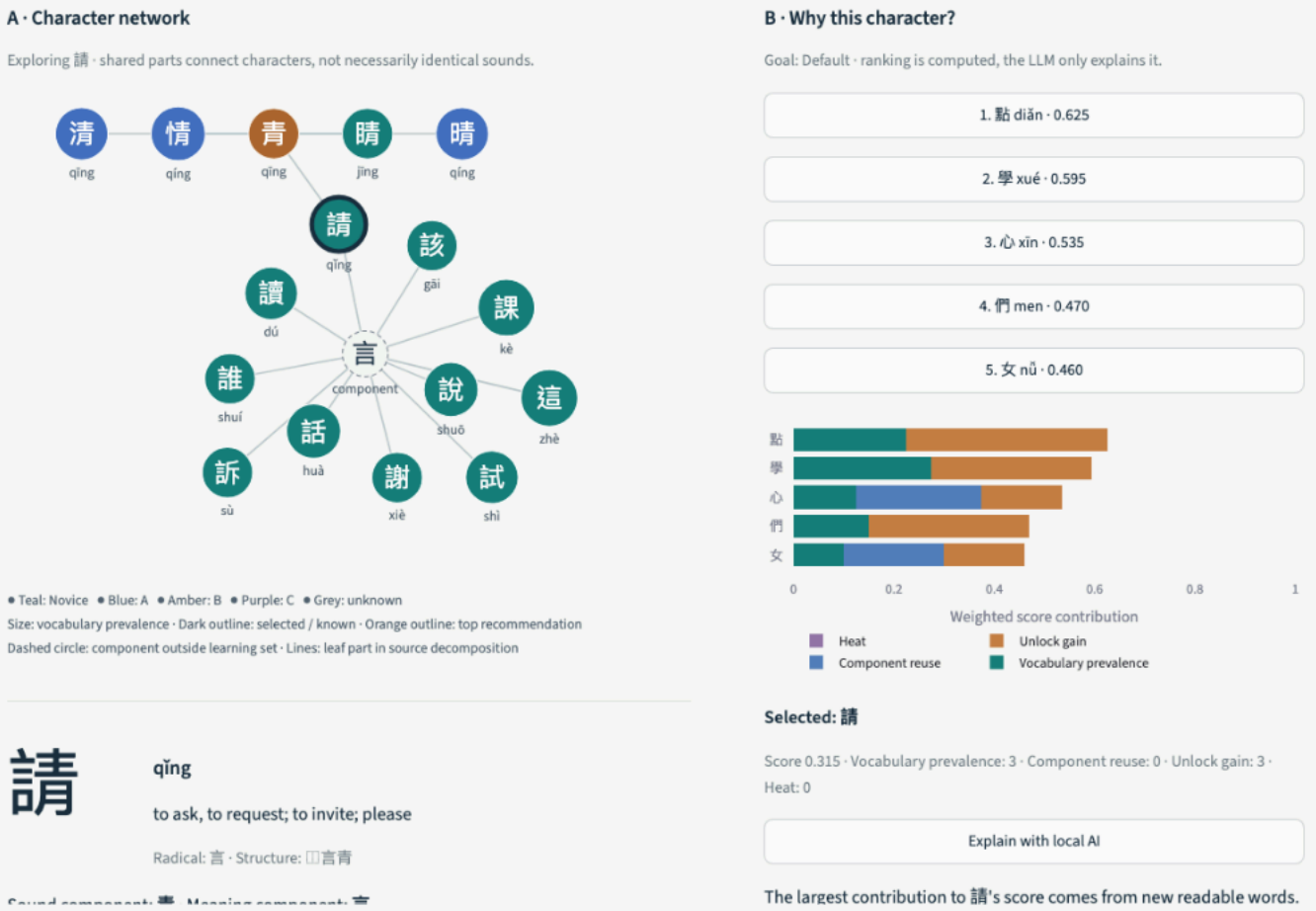
Unlock example

請問

Learning 請 unlocks this entry only when 問 is already known.

The LLM explains evidence. The numeric ranking comes from the displayed score.

Linked exploration and steering



Search and select

A graph click updates the selected character everywhere.

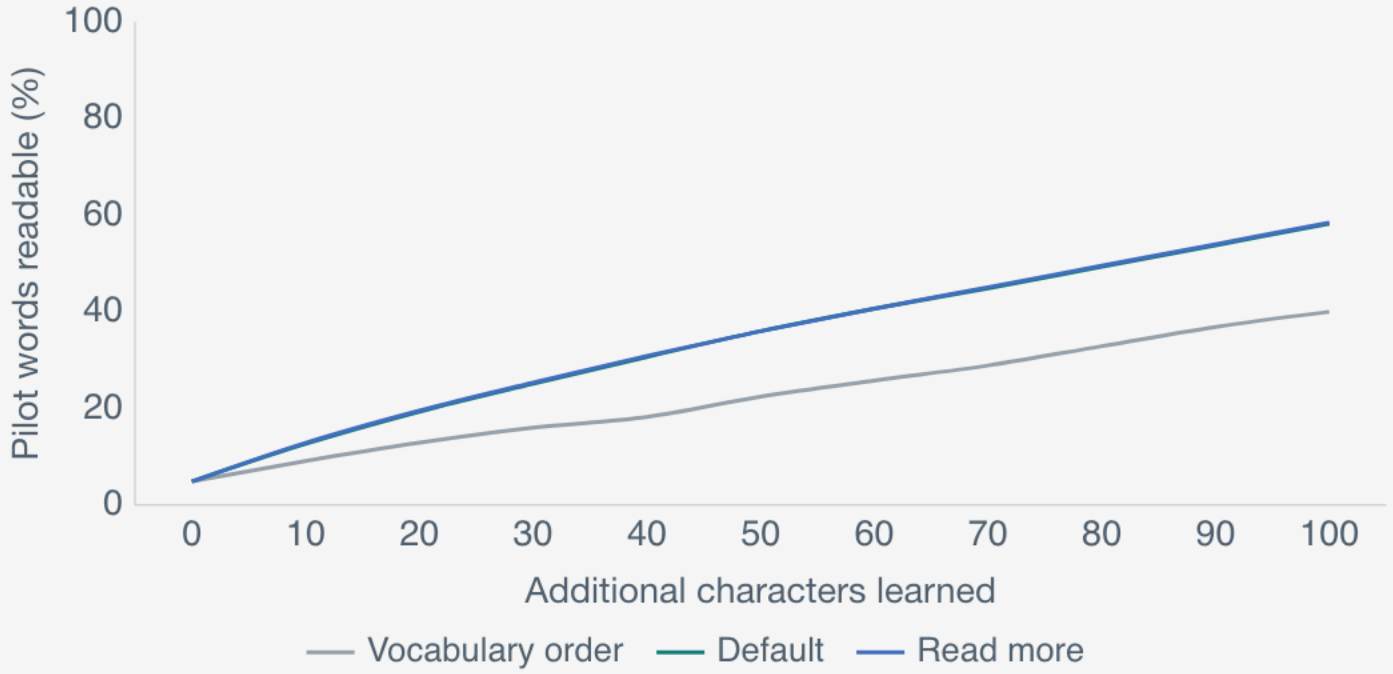
Steer the next step

Change the goal or weights.
Mark a character as known.
Recompute what becomes readable.

All screenshots show the running application.

Coverage comparison and validation

Same learner, same 450 words, 100 additional characters



14 automated tests passed

5 local LLM checks passed
Keyboard and responsive checks
No browser page errors

Next

Corpus frequency and richer data
Learner task testing
Language and usability review

In-sample vocabulary simulation. Learning effectiveness and real-text comprehension remain untested.